

## Week 1: Tilings — facilitator guide

For the three adults. Student pages: W1-K-v1 (grades K–1), W1-M-v1 (grades 2–3), W1-U-v1 (grades 4–5). Everything the children need to read is on their pages; everything else is here.

### 1. The session at a glance

- 0:00–0:05 Arrival, free handling. Pattern blocks are out on all three tables; the grades 4–5 table also has paper dominoes and its page 1 with the 4-by-4 boards. Children build anything they like.
- 0:05–0:08 Whole-group launch (organizer, section 3).
- 0:08–0:50 Table work, about forty minutes (sections 4–6).
- 0:50–0:57 Share (section 7).
- 0:57–1:00 Pieces back in the tubs, pages go home.

**The one question of the day.** *Can this shape be covered exactly? If not, how do you know for sure?* Every table meets the same theorem: on the triangle grid every blue rhombus covers one up-pointing and one down-pointing triangle, on the square grid every domino covers one dark and one light square, so a shape whose two counts differ cannot be covered. K–1 and grades 2–3 work with pattern blocks on the triangle grid; grades 4–5 work with paper dominoes and L-trominoes on the square grid. The three tables show each other the same argument at the end.

**Who sits where (a suggestion).** The parent takes K–1: section 4 scripts every problem and gives every answer, and the K–1 mathematics needs patience more than expertise. The second mathematician takes grades 2–3 (four children, and the hourglass board needs a push). The organizer takes grades 4–5 (the loop argument and the L-tromino induction). Any adult can run any table from this guide.

**Order everywhere: act, record, compare, explain.** The children build first, write what happened in the space on the page, look at several results side by side, and only then are asked why. Keep “how do you know for sure?” for the moment a child has three or four results in front of them.

### 2. Materials and preparation

**Printing.** Print the student pages at 100 % (no “fit to page”): the outlines are drawn at the true 1-inch edge of the pattern blocks, so a green triangle must fit a printed small triangle exactly. Check one page with a block before printing the rest. One copy per child plus one spare: K–1 pages  $\times 4$ , grades 2–3 pages  $\times 5$ , grades 4–5 pages  $\times 4$ . Three copies of this guide.

**Pattern blocks.** K–1 table (three children): at least 40 green triangles, 24 blue rhombi, 10 red trapezoids, 4 yellow hexagons, 4 purple chevrons. Grades 2–3 table (four children): at least 50 green, 32 blue, 12 red, 4 yellow, 4 purple. (The largest board, the side-4 triangle, takes 16 greens; a child covering it while a neighbor covers the side-3 triangle uses 25.) Grades 4–5 table: two blue rhombi only, for the launch.

**Paper pieces for grades 4–5.** Print the domino sheet (last pages of this guide) once and the L-tromino sheet once, on card stock if possible, and cut along the heavy lines before the session: 50 dominoes and 30 L-trominoes, that is 16 dominoes and 10 trominoes per child in two envelopes (two dominoes spare). The cell size (0.58 in) matches the boards of Problems 1, 2, 5, 6, 8, 9, 10, 11 and 13; the boards of Problems 7 and 12 are printed smaller and are drawn on, as are the three 8-by-8 boards, which need

more pieces than one child has.

**Also.** Pencils; one crayon or colored pencil per child (one color is enough: the coloring problems only distinguish colored from white); the launch card (last page) for the organizer; scissors in case a sheet was not cut.

### 3. Whole-group launch (three minutes)

Everyone stays at their table with the launch shape in front of them: for K–1 and grades 2–3 it is shape C on their page (the big triangle made of four small ones); the grades 4–5 table gets the launch card. Each table has two blue rhombi ready.

Say: “Every table has outlines and pieces today, and the job is always the same: cover the outline exactly. No gaps. No overlaps. Nothing sticking out past the line. Here is the first one: this triangle, and two blue rhombi. There is exactly enough blue: the triangle is four small triangles, and each rhombus is two. Cover it. Thirty seconds.”

Nobody will manage. Say: “Maybe we didn’t try hard enough. Or maybe it can’t be done. How could we know for sure, so sure that we would stop trying?” Take two answers, do not resolve them. Say: “That is today’s question at every table: which shapes can be covered, and when a shape can’t be, how do you know for sure. At the end each table will show one shape that cannot be covered and say why.” Tables begin.

### 4. Table K–1

**Shapes on the K–1 pages.** A: parallelogram (4 small triangles). B: red trapezoid outline (3). C: the launch triangle (4). D: yellow hexagon outline (6). E: side-3 triangle (9). F: parallelogram 2 by 2 (8). G: trapezoid with base 3 (8). H: side-4 triangle (16).

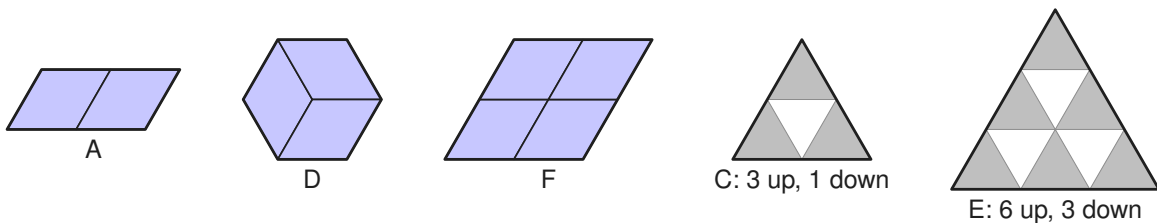
**Order and rough timing (40 minutes).** Problem 1 (4 min), 2 (7), 3 (5), 4 (3), 5 (6), 6 (5), 7 (4), 8 (4). That is the core; Problems 9–11 are for the rest of the hour. The adult reads each problem aloud, one sentence at a time, and scribes numbers when a child asks. The kindergartners will build and count; the first grader can write. Rotate who builds: while one child covers a shape, the other two are the checkers (“gaps? overlaps? sticking out?”).

**Problem 1** (hexagon: green, blue, red).

**Answer.** 6 green, 3 blue, 2 red. Insist on the three rules from the launch. If a child covers with a mix, say “Only green this time.” A child who cannot count six pieces reliably counts them off the shape one by one into your hand.

**Problems 2 and 3** (blue only; then green count).

**Answer.** A: check, 4 green. B: X, 3 green. C: X, 4 green. D: check (three blues meet at the center), 6 green. E: X, 9 green. F: check (four blues), 8 green.



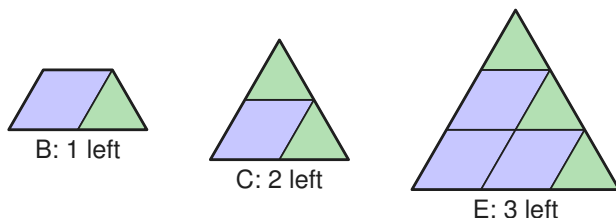
Let them try C for a full two minutes and try it yourself, visibly; the X should be earned. Do not explain yet. When a child says “it’s impossible”, ask “How sure are you? Sure enough to bet a cookie?” and move on; the rest of the hour is about earning that bet.

**Problem 4** (pairs).

**Answer.** B (3 triangles) and E (9) have one left over; circle B and E. Then point at C: “C has four, no leftover, and we still couldn’t cover it. So the number of triangles is not the whole story.” This is the whole reason for Problem 4: the counting rule (odd number, impossible) is true and easy, and C shows it is not enough.

**Problem 5** (trading). The child covers a shape with greens, then repeatedly swaps two side-by-side greens for one blue.

**Answer.** B: 1 green left. C: 2 left. E: 3 left. The leftover greens always point up, the same way as the big shape.



Ask “Which way do the leftover ones point? Why can’t two of them be traded?” (Two up-pointing triangles never touch along a side.) Keep the trading game to B, C and E: on the hexagon D an unlucky order of trades can leave two greens that do not touch, even though D can be covered, and that would confuse the point. The reason the trading always ends the same way on B, C, E: every blue is one up-triangle glued to one down-triangle, each trade uses up one down-triangle, and B, C, E have 1, 1, 3 down-triangles against 2, 3, 6 up-triangles. (The trading cannot stop early on these shapes: on B and C the single down-triangle touches every up-triangle, and on E each down-triangle touches a corner triangle that touches nothing else, so a green down-triangle always still has a green neighbor to trade with.)

**Problem 6** (coloring and the rhombus test). The child colors the 6 up-pointing triangles of the side-3 grid and tests a blue rhombus in five places.

**Answer.** Every row of the table reads 1 colored, 1 white. Ask “Can you find a place where it covers two colored ones?” They will try; there is none, because any two small triangles that share a side point opposite ways. Have a child say the rule in their own words: “a blue always takes one colored and one white.”

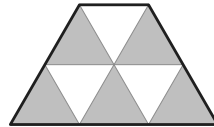
**Problem 7** (color and count C, D, F).

**Answer.** C: 3 colored, 1 white. D: 3 and 3. F: 4 and 4. Now put the results of Problems 2 and 7 side by side: D and F could be covered and have equal numbers; C could not and has 3 against 1. “Every blue needs one colored and one white. C has three colored and only one white. After one blue, two colored

ones are left with no white partner. That is how we know for sure.” This is the argument for the share; let the first grader rehearse it.

**Problem 8** (shape G: predict, then try).

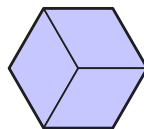
**Answer.** 5 colored, 3 white; guess X; trying confirms X. G has 8 triangles, four pairs, and still cannot be covered; the child has now predicted an impossibility before touching a piece. “You knew before you tried. That is what mathematicians do.”



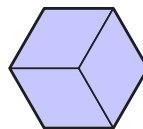
G: 5 up, 3 down

**Problem 9** (two ways for the hexagon, one for A).

**Answer.** Hexagon D: exactly two ways; each is the other turned by one sixth of a turn. Shape A: only one way. Why there is no third for D: every rhombus inside the hexagon touches the center point, so once the first rhombus is placed, the other two are forced; there are six possible rhombi and each way uses three, so there are two ways. Why A has only one: each sharp corner of A lies in a single small triangle, and that triangle has only one neighbor inside A, so the rhombus at each end is forced, and the two forced rhombi already fill A.



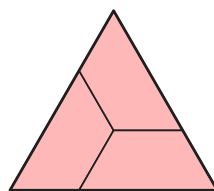
way 1



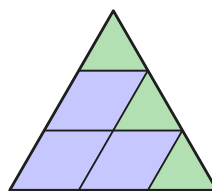
way 2

**Problem 10** (shape E with reds; fewest greens).

**Answer.** 3 red trapezoids, in a pinwheel (two mirror-image pinwheels exist). Fewest greens with blues and greens together: 3. Why not fewer: E has 6 up and 3 down triangles; each blue uses one down; after three blues no down is left, and the remaining up-triangles must be green. Three is also enough (picture).



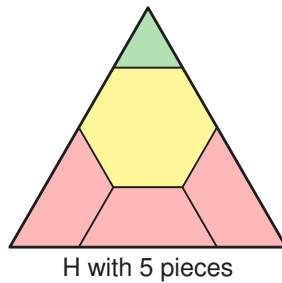
3 reds



3 blues + 3 greens

**Problem 11** (shape H with fewest pieces).

**Answer.** 5 pieces, for example one yellow hexagon, three red trapezoids and one green triangle (picture); an exhaustive search shows 4 is impossible. Why 4 is impossible, in a form a child can follow: H has 10 up-pointing and 6 down-pointing triangles, 4 more up than down. A blue, a yellow or a purple piece covers equally many of each, so it does not change the difference; a red or a green covers at most one more up than down. So at least 4 of the pieces must be red or green, and those 4 cover at most  $3 + 3 + 3 + 3 = 12$  triangles; the 4 or more triangles left need at least one more piece. That makes 5 at least, and 5 is enough (picture). (With the purple chevron allowed, an exhaustive search confirms that no 4 pieces work; keep the nonstandard pieces in the tub for this problem.)



**If the table stalls.** Pieces keep crossing the lines: switch to the shapes with small triangles drawn (Problems 5–7), where each piece has a home. A child insists C can be covered: hand over the two blues and watch together; then ask them to show you where the fourth triangle would go. Attention fades: make it a relay, one piece per child in turn. Counting slips: count pieces into your hand.

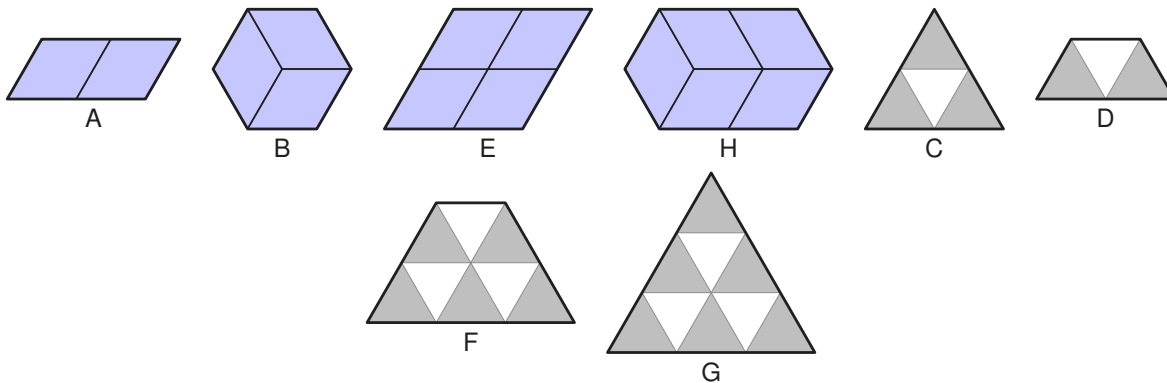
**If the table finishes early.** After Problem 11: “Cover H with red only.” (Impossible: 16 is not a multiple of 3; pairing three by three leaves one.) “Cover H with blue and green, fewest greens.” (4: H has 10 up and 6 down.) Then free building with a rule: “Build the biggest shape you can that blue alone can cover, and check it by coloring.”

## 5. Table grades 2–3

**Order and rough timing (40 minutes).** Problem 1 (7 min), 2 (6), 3 (3), 4 (5), 5 (5), 6 (5), 7 (6), 8 (6). Problems 9 and 10 are for children who finish. If time is short, do only board I in Problem 6 and only board L in Problem 7. Four children: pair them, two boards per pair, and have pairs swap results at Problem 3.

**Problems 1 and 2** (blue only, then green count).

**Answer.** A: check, 4. B: check, 6. C: X, 4. D: X, 3. E: check, 8. F: X, 8. G: X, 9. H: check, 10.



Do not explain the X's. Let a child who is sure say “impossible” and ask what would convince a skeptic.

**Problem 3** (table, odd numbers, same number different result).

**Answer.** Odd: D (3) and G (9). Same number, different result: A and C (both 4); E and F (both 8). The odd rule is correct (each rhombus covers two triangles) and is not enough: C and F have even numbers and still fail. Say so out loud; the rest of the page is about the finer reason.

**Problem 4** (coloring the side-4 grid; testing the pieces).

**Answer.** Blue: 1 colored, 1 white in all three directions. Red: 2 and 1, or 1 and 2. Green: 1 and 0, or 0

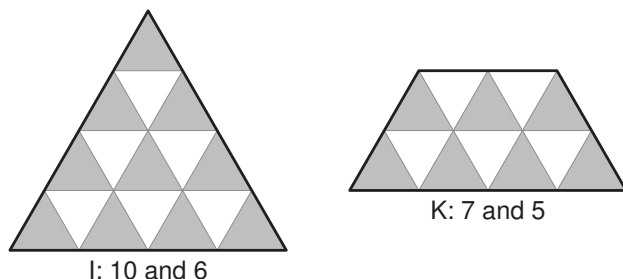
and 1. Yellow: 3 and 3. (A purple chevron: 2 and 2.) The fact that matters: a blue rhombus always covers one of each, because two small triangles sharing a side always point opposite ways.

**Problem 5** (color and count C, E, F; explain F).

**Answer.** C: 3 colored, 1 white. E: 4 and 4. F: 5 and 3. Explanation for F: four rhombi would cover four colored and four white triangles, but F has five colored and three white; so no four rhombi can cover F exactly. Accept any wording with the two ingredients: “each blue takes one of each” and “the counts are not equal.” A child who writes only “5 is not 3” should be asked “why does that matter for the rhombi?”

**Problem 6** (I and K: predict, then check).

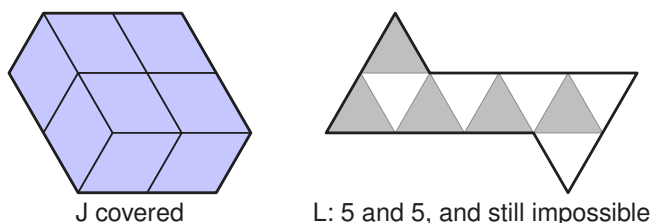
**Answer.** I: 10 colored, 6 white, prediction X, result X. K: 7 and 5, prediction X, result X.



A useful fact to notice with them: for a trapezoid the difference colored minus white equals the height in rows (F:  $5 - 3 = 2$ , K:  $7 - 5 = 2$ , the red piece:  $2 - 1 = 1$ ).

**Problem 7** (J and L).

**Answer.** J: 8 and 8, prediction check, result check (six coverings exist; one below). L: 5 and 5, prediction check, but the result is X. L is the important board of the day: equal counts are necessary but not sufficient.

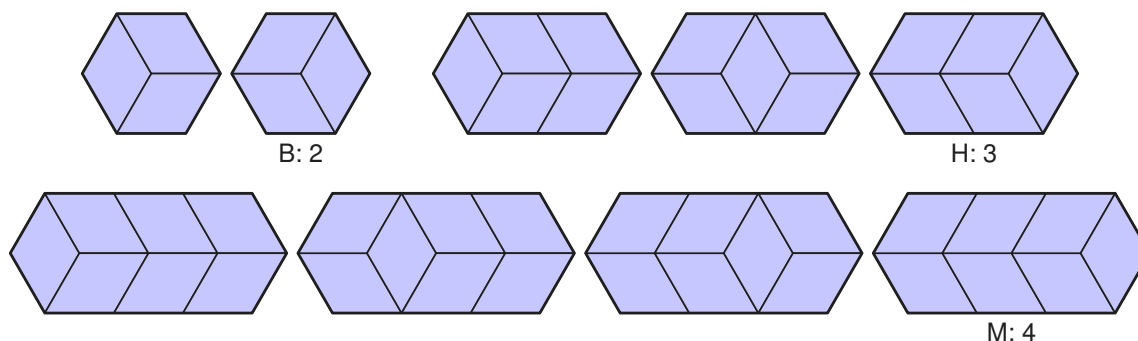


Why L fails: look at the left part alone, the big triangle of four small ones. It has 3 colored and 1 white. Only one rhombus can reach out of it into the rest of the shape, because the left triangle touches the rest along a single edge (the right edge of its bottom-right small triangle). Inside the left triangle at most one rhombus fits (it must use the one white), covering one colored; one more colored can leave through the neck; the third colored triangle has nowhere to go. So no covering exists, even though the whole shape has 5 and 5.

**Hints, in order.** (1) “Cover the left big triangle first. What happens every time?” (2) “How many rhombi can poke out of the left triangle into the rest, and where?” (3) “The left triangle has three colored and one white. One rhombus stays inside, one pokes out. What about the third colored one?” A child’s version of the argument is fine: “the left part is the launch triangle, and it can’t be done, and only one piece can escape.”

**Problem 8** (counting the ways).

**Answer.** B: 2 ways. H: 3 ways. M: 4 ways.

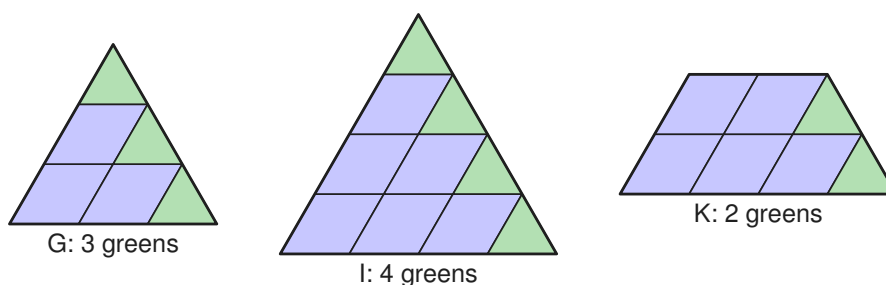


How to be sure there are no more: H is two rows of five small triangles. A rhombus either lies inside a row or stands across the two rows, and a standing rhombus can only stand on the 1st, 3rd or 5th triangle of the bottom row (the three that point down). Each row has an odd number of triangles, so at least one rhombus must stand; two standing rhombi would strand the triangle between them; so exactly one stands, in one of three places, and the rest of the covering is forced. M is two rows of seven, with four places for the standing rhombus: 4 ways. The pattern continues: a board like H with long side  $n$  has  $n + 1$  coverings. J has 6 (not for the children to count; the mathematician can say the number).

**Hints, in order.** (1) “Which rhombus in your covering is different from the others?” (2) “Where else could the standing one be?” (3) “Could two of them stand?”

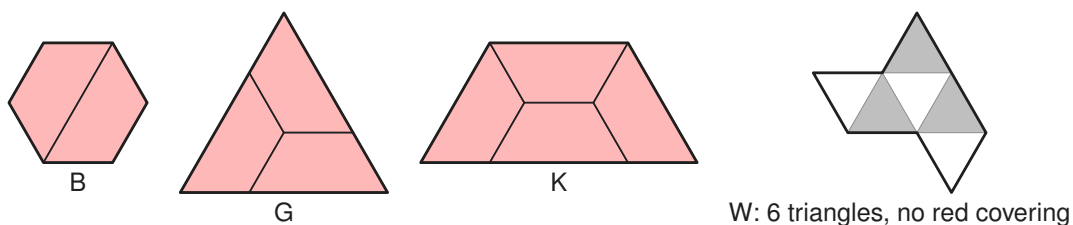
**Problem 9** (fewest greens).

**Answer.** G: 3. I: 4. K: 2. In every case the fewest is colored minus white: each blue removes one colored and one white, so after the whites are used up, the surplus colored triangles must be green; and that many greens always suffice on these boards (pictures).



**Problem 10** (red trapezoids only).

**Answer.** Check for B, D, G, K; X for A, C, E, F, H, I, J. Every board that can be covered has a number of triangles that is a multiple of 3 (6, 3, 9, 12).



The multiple-of-3 rule is necessary, not sufficient: the winged triangle W above (build it from six greens: the launch triangle with a down-triangle glued to two of its sides) has six triangles and no covering by trapezoids, because each wing can only be in a trapezoid that also uses the middle down-triangle, and there is only one middle. The up/down coloring gives no obstruction for trapezoids: a red piece covers

2 and 1 or 1 and 2, so it changes the difference by 1 either way. (Other weightings can: on  $W$ , give the middle down-triangle weight 1 and the rest 0; every trapezoid inside  $W$  contains it, so two trapezoids would need total weight 2, but  $W$  has 1.)

**If the table stalls.** On Problem 1, a pair that “finds” a covering of  $C$  or  $F$  has a piece over a line; check with a finger along the outline. On Problem 5, a child who cannot start the explanation should first say out loud what a rhombus covers, then count  $F$ ’s colors, then put the two sentences together. On Problem 7, use the hints above; if  $L$  is still stuck after hint 3, give the argument yourself and ask them to repeat it on the shape.

**If the table finishes early.** Problems 9–10; then: (a) a hexagon like  $H$  with long side 4 has 5 coverings (draw it on triangle paper, or argue it from the standing rhombus); (b) the side-2 hexagon  $N$  (last page of this guide, printed at true scale) can be covered by 12 blues in 20 ways; ask for three different coverings, then to color each covering with three colors, one per rhombus direction: the picture becomes a stack of cubes, and different coverings are different stacks; (c) “build a shape with equal counts that blue cannot cover” (the hourglass idea again, with a different neck).

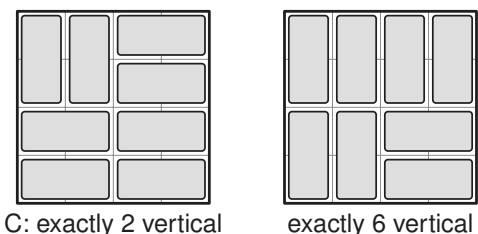
## 6. Table grades 4–5

**Conventions.** Rows are counted from the top, columns from the left; “dark” means the color of the top-left square. An 8-by-8 board has 32 dark and 32 light squares; a 4-by-4 has 8 and 8; a 3-by-3 has 5 dark (the corners and the center) and 4 light; a 5-by-5 has 13 dark and 12 light.

**Order and rough timing (40 minutes).** Problem 1 (4 min), 2 (6), 3 (3), 4 (3), 5 (6), 6 (3), 7 (4), 8 (6), 10 (4), 11 (5). That is the core: the coloring argument, the loop argument, and the tromino induction. Problem 9 (drawing the loop on the 8-by-8) is skipped if time is short and offered at the end; Problems 12 and 13 are for the rest of the hour. Children draw their tilings on the boards as they go: the drawing is the record.

**Problem 1** (4-by-4 warm-up).

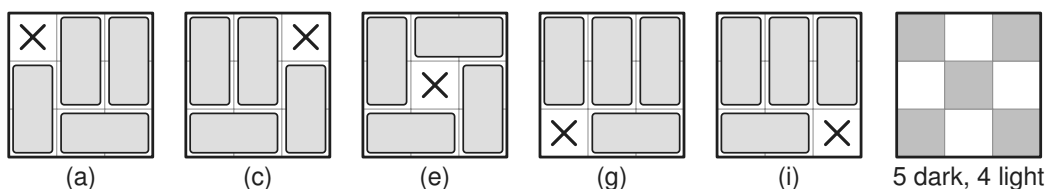
**Answer.** Any covering of  $A$  and  $B$  (there are 36 in all).  $C$ : a covering with exactly 2 vertical dominoes exists (picture).  $D$ : exactly 3 vertical dominoes is impossible; whatever the children end up with, every number of vertical dominoes written on their lines is even.



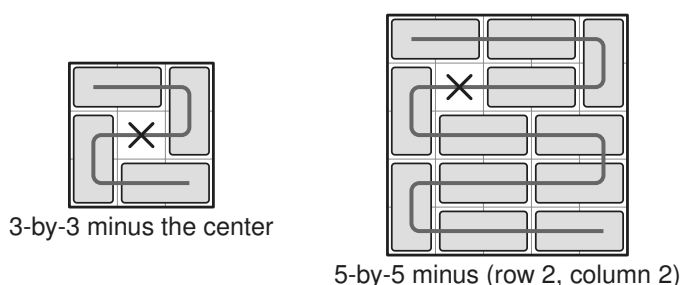
Why  $D$  is impossible: in any row, the horizontal dominoes cover an even number of squares, so the number of squares covered by vertical dominoes in that row is even (4 minus even). Every vertical domino in rows 1–2 has a square in row 1, so their number is even. Every vertical domino touching row 2 lies in rows 1–2 or rows 2–3, and the total is even, so the number in rows 2–3 is even; likewise rows 3–4. The number of vertical dominoes is a sum of three even numbers. So 0, 2, 4, 6, 8 are possible (2 and 6 above; 0, 4 and 8 with rows or columns) and 3 is not. Do not spend more than a minute on  $D$  at the start; children who insist can come back to it.

**Problem 2** (3-by-3 minus one square).

**Answer.** Coverable: (a), (c), (e), (g), (i), the corners and the center. Impossible: (b), (d), (f), (h), the middles of the sides.



When the nine boxes are filled, the picture of checks and X's is itself a checkerboard. Point at it and say nothing yet. For the adult only (it comes back in Problem 8): why every dark square works. Walk the snake path through all nine squares; the colors alternate along it and it starts and ends dark, so removing a dark square cuts it into two pieces with an even number of squares each, and dominoes follow the path. The same snake works on the 5-by-5 and 7-by-7 boards.



**Problem 3** (checkerboard, domino test).

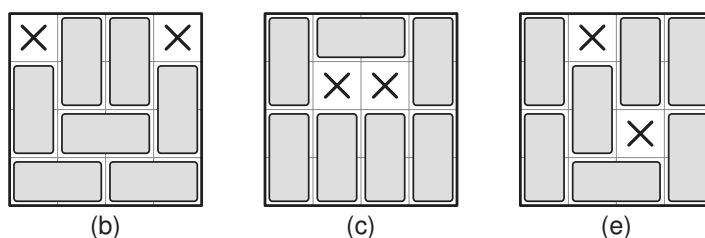
**Answer.** 5 dark, 4 light. A domino covers 1 dark and 1 light in every position. Two squares that share a side always have different colors.

**Problem 4** (explain the X's).

**Answer.** A side-middle square is light. Crossing it out leaves 5 dark and 3 light. Four dominoes cover 4 dark and 4 light, so they cannot cover this board. Crossing out a corner or the center (dark) leaves 4 and 4, and those boards can indeed be covered. The two ingredients must both appear: what one domino covers, and the counts.

**Problem 5** (4-by-4 minus two squares).

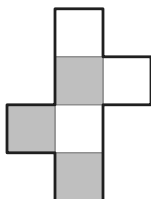
**Answer.** (a) 6 dark, 8 light, X. (b) 7 and 7, check. (c) 7 and 7, check. (d) 6 and 8, X. (e) 7 and 7, check. (f) 8 and 6, X. Coverings for (b), (c), (e):



The prediction rule “different colors: possible; same color: impossible” is right on all six, and it is right on every pair of squares of the 4-by-4 (Problem 8 says why). A child who predicts from the counts and is right three times in a row has learned the theorem.

**Problem 6** (board S).

**Answer.** 3 dark, 3 light, and still X. Why: the top square touches only one other square, the one below it; the right square of the second row also touches only that same square. Both need it, and a domino covering it can serve only one of them. (The same happens at the bottom.) Equal counts are necessary, not sufficient.

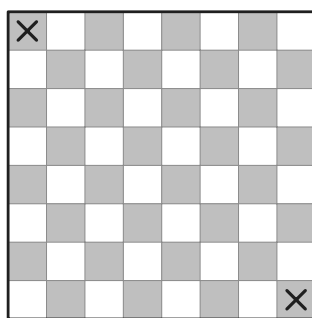


S: two squares that both need the same neighbor

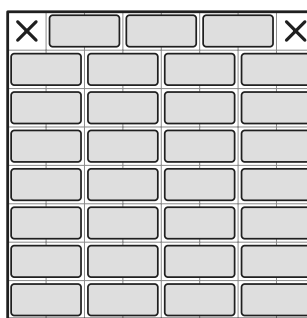
**Hints, in order.** (1) “Which squares of S touch only one other square?” (2) “What do those squares both want?”

**Problem 7** (T and U without pieces).

**Answer.** T: X. The two opposite corners have the same color (both dark): 30 dark and 32 light are left, and 31 dominoes would cover 31 and 31. U: check. The corners on one side have different colors, 31 and 31 remain, and a covering is easy: the top row has six squares (three horizontal dominoes) and each other row has eight (four).



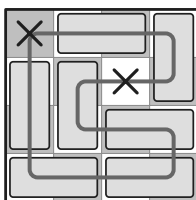
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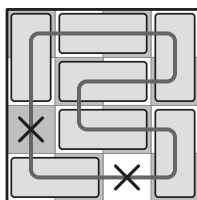
U

**Problem 8** (the loop). The path on V and W visits all 16 squares and returns to its start. Cross out one dark and one light square; the loop falls into two pieces, and dominoes are laid along each piece.

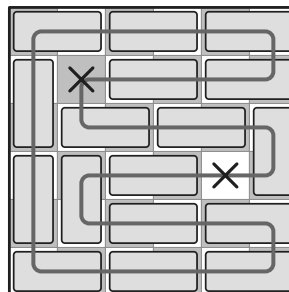
**Answer.** Any choice works. The two examples below cross out (row 1, column 1) with (row 2, column 3), and (row 3, column 1) with (row 4, column 3).



V



W



X, with the comb loop

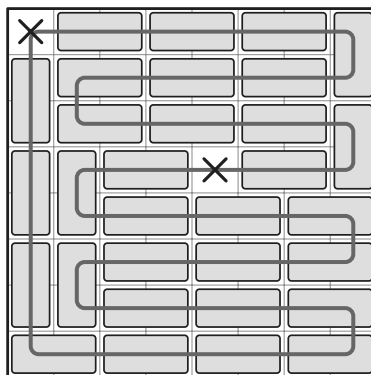
For X the children draw their own loop; the comb above (go up and down the columns, come back along the first column) is the easiest, and it works on any board with an even number of rows. Why the method

always works: walking around the loop, the colors alternate dark, light, dark, light. Cut out one dark and one light square. Each remaining piece starts just after one of the crossed squares and ends just before the other; a piece that starts after the dark square starts light and ends dark, so it has an even number of squares, and the dominoes pair them up along the piece. The other piece is the same with the colors swapped. If the two crossed squares are neighbors on the loop, one piece is empty and the other has 14 squares.

**Hints, in order.** (1) “Walk around the loop and say the colors.” (2) “What color is the first square after the crossed dark one? The last square before the crossed light one?” (3) “So how many squares are in that piece: even or odd?” This is Gomory’s theorem: an 8-by-8 board with any one dark and any one light square removed can always be covered by dominoes.

**Problem 9 (Y).**

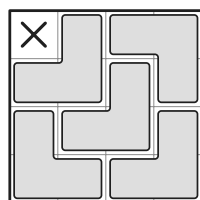
**Answer.** Draw the comb loop through all 64 squares; the crossed squares have different colors (row 1, column 1 is dark; row 4, column 5 is light), so the loop method covers the rest. If both crossed squares were dark, the two pieces of the loop would start light and end light, each with an odd number of squares, and the method fails; indeed no covering exists, by the count 30 dark against 32 light.



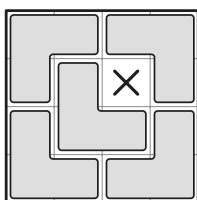
Y along the comb loop

**Problem 10 (4-by-4 minus one square, L-trominoes).**

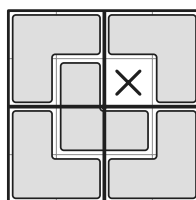
**Answer.** Every square works: there is no bad choice. By symmetry there are only three kinds of square (corner, side, inner), and all three can be done.



(a)



(b)

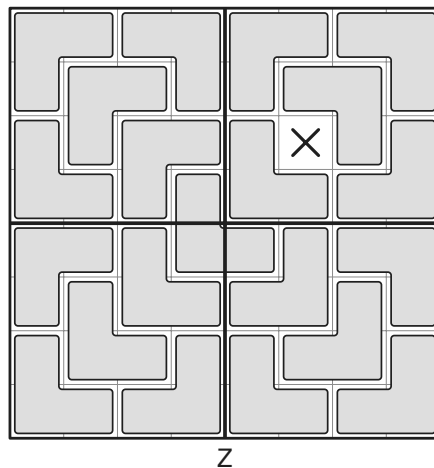


(b) by the plan of Problem 11

The reason every square works is the plan of Problem 11 applied to the 4-by-4: split into four 2-by-2 blocks; the crossed square is in one block; put one tromino at the center covering one square of each of the other three blocks; now each block is a 2-by-2 with one square missing, which is exactly one tromino.

**Problem 11 (8-by-8 minus one square, the induction).**

**Answer.** The crossed square (row 3, column 6) lies in the top-right quarter. The first tromino sits at the center and covers the three central squares of the other quarters. Each quarter is now a 4-by-4 with one square missing, and the plan of Problem 10 finishes each quarter. 21 trominoes in all (64 minus 1 is 63, and 63 divided by 3 is 21).

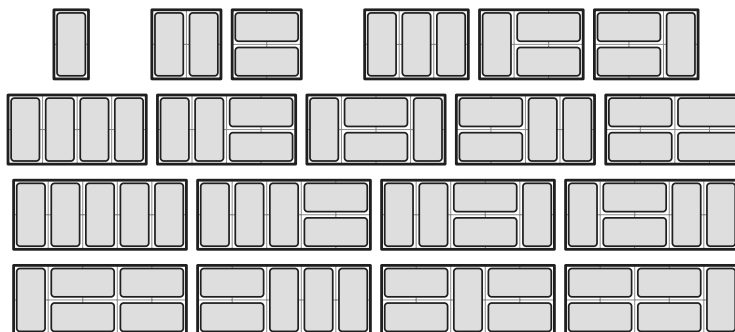


For a 16-by-16 board with one square crossed out: split it into four 8-by-8 quarters, place one tromino at the center covering one square in each of the three quarters without the cross, and each quarter is now an 8-by-8 with one square missing, which is Problem 11 again. It takes  $(256 - 1)/3 = 85$  trominoes; in general a  $2^n$ -by- $2^n$  board with one square missing takes  $(4^n - 1)/3$ . The point to make out loud: the child never has to draw the 16-by-16; knowing how to do 8 from 4 is knowing how to do 16 from 8. That is a proof by induction.

**Hints, in order.** (1) “Which quarter is the cross in?” (2) “Put one tromino at the very center so that it stays out of that quarter.” (3) “Now look at one quarter alone. What problem is it?”

**Problem 12** (2-by- $n$  counting).

**Answer.** 1, 2, 3, 5, 8 ways; the 2-by-6 has 13. All of them:

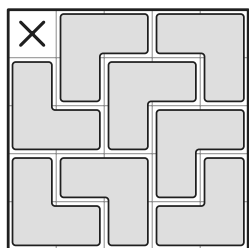


How to be sure every way was found: look at the left end of the board. Either one vertical domino stands there, and the rest is a 2-by- $(n - 1)$  board, or two horizontal dominoes lie there one above the other, and the rest is a 2-by- $(n - 2)$  board. Nothing else is possible at the left end, and the two cases never overlap. So the number of ways for  $n$  is the number for  $n - 1$  plus the number for  $n - 2$ :  $5 = 3 + 2$ ,  $8 = 5 + 3$ ,  $13 = 8 + 5$ . (These are the Fibonacci numbers.)

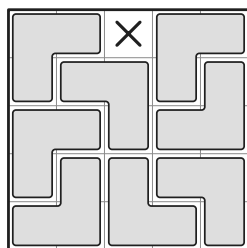
**Hints, in order.** (1) “What can the left end of the board look like?” (2) “After you decide the left end, what board is left?”

**Problem 13** (5-by-5 minus one square, L-trominoes).

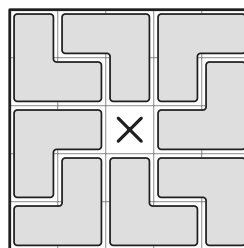
**Answer.** (a) check, (b) X, (c) check, (d) X. Map: exactly the nine squares whose row and column are both odd (1, 3 or 5), that is the four corners, the four middles of the sides and the center, can be crossed out; the other sixteen cannot.



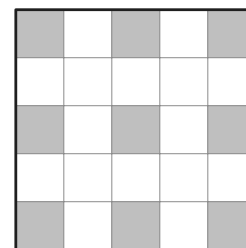
(a)



side-middle, row 1, column 3 (not (b))



(c)



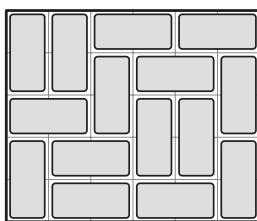
map: shaded squares work

Why the other squares fail: mark the nine squares with both coordinates odd (shaded on the map). Any 2-by-2 block of the board contains exactly one marked square, because among two consecutive rows exactly one is odd, and likewise for columns. An L-tromino lies inside a 2-by-2 block and covers three of its four squares, so it covers at most one marked square. Eight trominoes cover at most eight marked squares. If the crossed-out square is unmarked, all nine marked squares must be covered: impossible. Why the marked squares all work: by the symmetries of the board there are only three kinds (corner, side-middle, center), and the three pictures above do them.

**Hints, in order.** (1) “Shade the squares with an odd row and an odd column. How many are there?” (2) “How many shaded squares can one tromino cover?” (3) “So how many can eight cover?”

**If the table stalls.** Problem 2: a child “covering” a side-middle board has a domino over the crossed square or the edge. Problem 4: ask for the two sentences separately (what a domino covers; the counts) and then together. Problem 8: walk the loop with a finger and say the colors aloud together. Problem 11: do the first tromino for them, then hand over.

**If the table finishes early.** Problems 12–13; then, in this order: (a) Problem 1 board D: ask why every number of vertical dominoes on the lines is even, then give the parity argument from Problem 1; (b) fault-free coverings: cover a 6-by-6 board with dominoes so that no straight line cuts across the whole board without passing through a domino. It is impossible: each of the 10 inside lines must be crossed by at least one domino; the part of the board on one side of a line has an even number of squares, the dominoes not crossing the line contribute 0 or 2 squares to it, so the number of dominoes crossing the line is even, hence at least 2; that is at least 20 crossings, but each domino crosses exactly one line and there are only 18 dominoes. A 5-by-6 board can be done:



(c) counting 3-by- $n$  coverings: 3-by-2 has 3, 3-by-4 has 11, 3-by-6 has 41; (d) 7-by-7 with one square crossed out and L-trominoes: every square works (checked by computer), and a child can look for the plan that does it.

## 7. The share (seven minutes)

Each table shows one shape that cannot be covered and gives its reason; one child speaks, the adult holds the shape.

- K–1: shape C colored (3 colored, 1 white) with two blue rhombi. “A blue always takes one colored and one white. There is only one white.”
- Grades 2–3: board F (5 and 3) with blue rhombi, and if there is time, board L: “equal numbers, and still impossible, because of the neck.”
- Grades 4–5: board T, the 8-by-8 with opposite corners, and the checkerboard sentence: “both corners are dark; 30 dark, 32 light; every domino takes one of each.”

The organizer closes: “Three tables, three kinds of pieces, one reason: every piece takes one of each color, so the two colors must match. And two tables found shapes where the colors match and it still can’t be done, so mathematicians keep looking for better reasons. Next week we keep the pieces and change the question.” If a minute remains, grades 2–3 show the two coverings of the hexagon and grades 4–5 show 1, 2, 3, 5, 8.

## 8. The mathematics behind the session (for the two mathematicians)

**Tilings are perfect matchings.** Put a vertex in each cell and join cells that share an edge. On both grids the graph is bipartite: squares by checkerboard color, triangles by orientation. A domino or rhombus tiling is a perfect matching, so “can it be tiled?” is answered by Hall’s theorem: the two classes must be equal and every set  $S$  of dark cells must have at least  $|S|$  light neighbors. The color count is Hall’s condition for  $S$  = all dark cells; board S and the hourglass L violate it for a small  $S$  (two cells with one common neighbor; a sub-triangle with a surplus of two and one exit). Matching is polynomial (Hopcroft–Karp), and for simply connected regions Thurston’s height functions give a linear-time test and an explicit obstruction. The K–1 trading game is a greedy matching, which is why it can get stuck on the hexagon: the fix is an augmenting path.

**Coloring arguments and their limits.** A coloring argument assigns weights to cells so that every tile has the same total: up minus down for rhombi, dark minus light for dominoes, “both coordinates odd” for the 5-by-5 tromino problem, where every L covers at most one weighted cell. Conway and Lagarias (1990) and Thurston (1990) showed that some impossibility results, for three-cell tiles on triangular regions of the hexagonal grid, cannot be proved by any coloring and need the tiling group, or the height function. That is where “how do you know for sure?” leads.

**Counting.** 2-by- $n$  boards: Fibonacci, by the transfer argument of Problem 12. Rectangles: Kasteleyn (1961) and Temperley and Fisher (1961) gave a closed product formula; the 4-by-4 has 36 domino tilings, the 6-by-6 has 6,728, the 8-by-8 has 12,988,816. Lozenge tilings of the hexagon with sides  $a, b, c$  are plane partitions in an  $a \times b \times c$  box, counted by MacMahon’s product  $\prod_{i,j,k} \frac{i+j+k-1}{i+j+k-2}$ ; boards H and M are the case  $b = c = 1$  with  $a = 2, 3$ , giving  $a + 1$ ; the side-2 hexagon N has 20. Drawn with one shade per rhombus direction, a lozenge tiling is a stack of unit cubes seen from a corner; the two tilings of the unit hexagon are a cube seen from outside and from inside.

**Reconfiguration.** Any two domino tilings of a simply connected region are joined by a sequence of 2-by-2 flips (Thurston), and any two lozenge tilings by hexagon flips; the two coverings of board B are one flip. Flips drive the Markov chains used to sample random tilings, and random tilings of large regions show frozen and disordered zones separated by a curve: the arctic circle of the Aztec diamond (Jockusch, Propp, Shor) and the inscribed ellipse of the hexagon (Cohn, Larsen, Propp). This is the dimer model of statistical mechanics.

**L-trominoes.** Golomb (1954) proved the  $2^n$  induction of Problem 11. Chu and Johnsonbaugh (1986) settled all  $n$ -by- $n$  boards with one square removed: possible for every square when  $3 \nmid n$  and  $n \neq 5$ , never

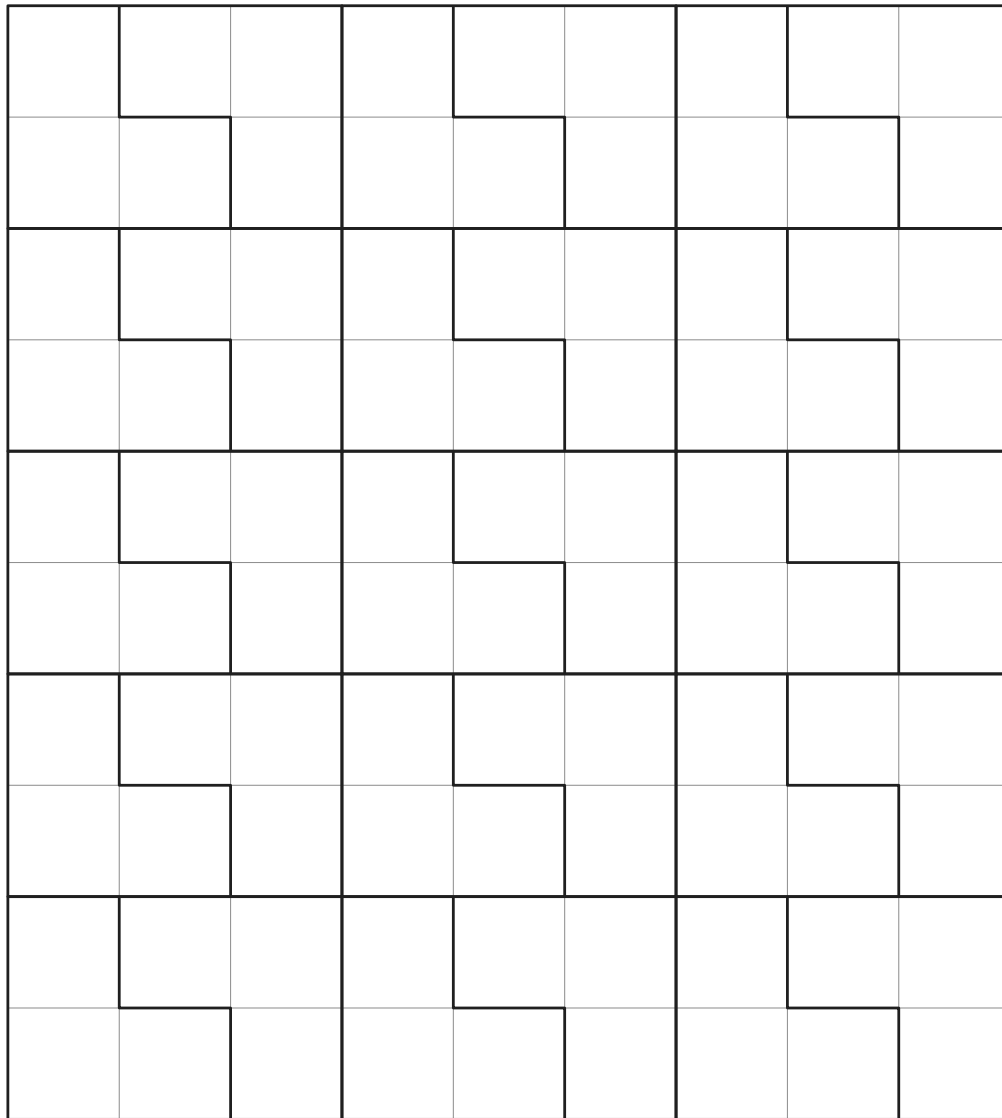
when  $3 \nmid n$  (the area is not a multiple of 3), and for  $n = 5$  exactly the nine squares of Problem 13. Tiling an arbitrary region with L-trominoes is NP-complete (Moore and Robson, 2001), and Berger (1966) showed that whether a finite set of tiles can tile the whole plane is undecidable: tileability is easy for dominoes and hard in general.

**Gomory's theorem** (Problem 8) is usually stated for the 8-by-8 board: remove any one dark and any one light square and the rest can be tiled. The Hamiltonian cycle proof is the one the children do; the same idea with a Hamiltonian path (the snake pictures under Problem 2) shows that an odd-by-odd board minus any square of the majority color can be tiled.

**Appendix A. Domino sheet for grades 4–5 (print once, cut along the heavy lines: 50 dominoes)**

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**Appendix B. L-tromino sheet for grades 4–5 (print once, cut along the heavy lines: 30 trominoes)**



## Appendix C. Launch card, and extra board N for grades 2–3 (true 1-inch scale)

The launch card is the triangle of side 2: four small triangles, two blue rhombi. Board N is the hexagon of side 2: 24 small triangles, 12 blue rhombi, 20 different coverings.

